

Question Paper

Exam Date & Time: 18-Jun-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

FOURTH SEMESTER B.TECH. DEGREE EXAMINATIONS - JUNE 2025

SUBJECT: CSE 2221 / CSE_2254 - FORMAL LANGUAGES AND AUTOMATA THEORY

(COMPUTER SCIENCE / COMPUTER SCIENCE AND ENGINEERING - CYBER SECURITY / COMPUTER SCIENCE AND ENGINEERING - ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING / COMPUTER SCIENCE AND ENGINEERING (DATA SCIENCE))
(MAKE-UP)

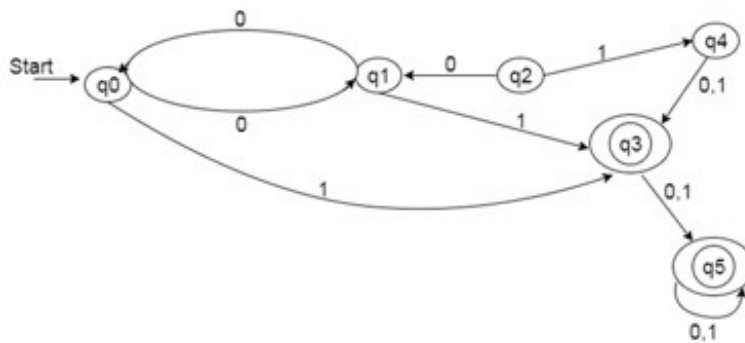
Marks: 50

Duration: 180 mins.

Answer all the questions.

- 1A) Let $L = \{w \in \{a, *, /\}^* \mid w \text{ represents a C-style comment}\}$. Design a Deterministic Finite Automata (DFA) in the form of a transition diagram. Identify some strings that may be accepted and rejected by this language. (4)

- 1B) Minimize the DFA given below. (4)



- 1C) Construct a context-sensitive grammar for the language $L = \{a^n b^n c^n / n \geq 1\}$, where the number of a's, b's, and c's must be equal. (2)
- 2A) Give one example of a problem that cannot be solved by a Turing Machine, and explain why it is undecidable. (2)
- 2B) Given the language $L = \{w \in \{a, b\}^* \mid \text{the number of a's in } w \text{ is even and the number of b's is odd}\}$, find a regular expression for L. Then, construct a DFA that accepts the same language. (3)
- 2C) (5)

Obtain the string 'aaabbabbba' by applying leftmost derivation and the parse tree for the grammar shown below. Is it possible to obtain the same string again by applying leftmost derivation but by selecting different productions?

$S \rightarrow aB|bA$
 $A \rightarrow aS|bAA|a$
 $B \rightarrow bS|aBB|b$

Write the regular expression for the above grammar.

3A) Construct λ -NFA for the regular expression given below: (3)

- i. $((ab+bc)(a+bc)^*)^*$
- ii. $(b+bc+ca)(a+b+c)(abc)^*$

3B) Find context-free grammar for the language (3)

$L = \{a^n b^m c^k \mid k = m \bmod 3 \text{ and } n \geq 0, m \geq 0\}$.

3C) Given a grammar $G = (\{S, A, B, C\}, \{a, b\}, S, P)$ with P given as : (4)
 $S \rightarrow aA \mid aBB$, $A \rightarrow aaA \mid \lambda$, $B \rightarrow bB \mid bbC$, $C \rightarrow B$, eliminate all useless productions and then describe the language generated by the resulting simplified grammar.

4A) Is the PDA to accept the language $L(M) = \{w C w^R \mid w \in (a,b)^*\}$ deterministic ? Show all the transitions defined for the language and justify your answer. (3)

4B) Illustrate with a proper example, how CFLs are not closed under intersection and complementation. (3)

4C) Construct a Transducer using state transition diagram to reverse the given string $w \in \{0,1\}^*$ such that the final state halts at the beginning of the reversed string. (4)

5A) Design Turing Machine using state transition diagram to accept all even and odd palindromes on $\Sigma = \{a, b\}$. (4)

5B) Find an unrestricted grammar to generate a language (4)

$L = \{w \in \{a, b, c\}^* \mid n_a(w) = n_b(w) = n_c(w)\}$. Show the steps to derive a string $w = \{aabbcc\}$.

5C) Find the language generated by the grammar (2)

$G = (\{S, T\}, \{a, b\}, S, P)$ where the set of productions rule P is given as $S \rightarrow a a T / a T a / T a a$, $T \rightarrow a T b \mid b T a \mid a T a \mid b T b \mid \epsilon$.

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Event Name		Subject	
MIT MPL and BLR- BTech-MSc-MCA - II-IV and VI Sem - Makeup Examination - Jun 2025		CSE-2254: Formal Languages and Automata Theory	
#	Course Outcome	Description	Marks

No records found / CO Level Description not defined

* **NA** - CO Not Selected

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#	Level	No of Questions	Marks (%)
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No records found / Blooms Level Description not defined